Reg. No. | | | | | | | | | | |

[13CE51]

FIFTH SEMESTER B.TECH/ B.E EXAMINATION, DECEMBER -2015

CONCRETE TECHNOLOGY

Time:3 Hours **Maximum Marks: 70**

Instructions: 1. Answer Any FIVEfull Questions
2. Use of IS 10262:2009 and IS456 - are permitted.

1. a. Explain heat of hydration of cement. **06**
   b. Explain setting time and soundness of cement with BIS specifications. **08**

2. a. What is meant by grading of aggregate? Explain the importance of size, shape and texture with respect to coarse aggregate. **07**
   b. Explain bulking of sand and its importance? **07**

3. a. Define workability? What are the factors affecting workability? Explain any two factors affecting workability of fresh concrete. **08**
   b. What are the factors affecting strength of concrete? **06**

4. a. Why is curing necessary? Explain briefly the different types of curing. **08**
   b. Write a note on: **06**
   i. Batching
   ii. Mixing

5. a. What is Ready Mix Concrete? Mention the merits and demerits of RMC. **07**
   b. What is light weight concrete? What are the properties and uses of Light Weight concrete? **07**

6. Design the concrete mix for M_{20} grade concrete with the following data: **14**
   i. Characteristic compressive strength at 28 days – 20Mpa.
   ii. Maximum size of aggregate – 20mm.
   iii. Design of workability compaction factor-0.8 (C.F).
   iv. Degree of quality control-Good.
   iv. Type of exposure: Mild.
   v. Sp.gravity of Five aggregate=2.60
   Sp.gravity of coarse aggregate=2.60
   Sp.gravity of cement=3.15.
   Assume any other data required.

7. a. Write a short note on: **07**
   i. Silica fume
   ii. GGBFS
   b. Explain the role of Super plasticizers in modifying the properties of concrete. **07**

8. a. Explain Sulphate attack. **06**
   b. Explain the Mechanism involved in electro chemical corrosion. **08**

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